PCET Education Trust

Pimpri Chinchwad College OF Engineering, Pune

Database Management Systems Lab

ASSIGNMENT NO. 02

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Batch A1

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Case Study: Airline Management System

SQL Queries:

1. SHOW: Displays the list of databases or tables available in the MySQL server or selected database.

```
mysql> show databases;
+--------------------+
| Database           |
+--------------------+
| airline            |
| employee           |
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
6 rows in set (0.00 sec)
```

```
mysql> SHOW TABLES;
+-------------------+
| Tables_in_airline |
+-------------------+
| aircraft          |
| airports          |
| bookings          |
| crew              |
| customers         |
| flights           |
+-------------------+
6 rows in set (0.03 sec)
```

2. USE: Selects a database to work with for executing SQL queries.

```
mysql> use Airline;
Database changed
```

3. CREATE: Used to create a new database or table in MySQL.

```
mysql> create database Airline;
Query OK, 1 row affected (0.02 sec)
```

```
mysql> CREATE TABLE Airports (
    ->     airport_id INT PRIMARY KEY AUTO_INCREMENT,
    ->     name VARCHAR(100),
    ->     city VARCHAR(50),
    ->     country VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.21 sec)

mysql> CREATE TABLE Aircraft (
    ->     aircraft_id INT PRIMARY KEY AUTO_INCREMENT,
    ->     model VARCHAR(50),
    ->     capacity INT
    -> );
Query OK, 0 rows affected (0.06 sec)

mysql> CREATE TABLE Flights (
    ->     flight_id INT PRIMARY KEY AUTO_INCREMENT,
    ->     flight_number VARCHAR(10),
    ->     source_airport INT,
    ->     destination_airport INT,
    ->     departure_time DATETIME,
    ->     arrival_time DATETIME,
    ->     aircraft_id INT,
    ->
    ->     FOREIGN KEY (source_airport) REFERENCES Airports(airport_id),
    ->     FOREIGN KEY (destination_airport) REFERENCES Airports(airport_id),
    ->     FOREIGN KEY (aircraft_id) REFERENCES Aircraft(aircraft_id)
    -> );
Query OK, 0 rows affected (0.10 sec)

mysql> CREATE TABLE Passengers (
    ->     passenger_id INT PRIMARY KEY AUTO_INCREMENT,
    ->     name VARCHAR(100),
    ->     email VARCHAR(100),
    ->     phone VARCHAR(15)
    -> );
Query OK, 0 rows affected (0.06 sec)
```

4. DESC (DESCRIBE): Displays the structure of a table including its columns, data types, and constraints.

```
mysql> DESC Passengers;
+--------------+--------------+------+-----+---------+----------------+
| Field        | Type         | Null | Key | Default | Extra          |
+--------------+--------------+------+-----+---------+----------------+
| passenger_id | int          | NO   | PRI | NULL    | auto_increment |
| name         | varchar(100) | YES  |     | NULL    |                |
| email        | varchar(100) | YES  |     | NULL    |                |
| phone        | varchar(15)  | YES  |     | NULL    |                |
+--------------+--------------+------+-----+---------+----------------+
4 rows in set (0.10 sec)
```

5. INSERT: Adds new records (rows) into a table.

```
mysql> INSERT INTO Airports (name, city, country) VALUES
    -> ('Indira Gandhi International Airport', 'Delhi', 'India'),
    -> ('Chhatrapati Shivaji Maharaj International Airport', 'Mumbai', 'India'),
    -> ('Kempegowda International Airport', 'Bangalore', 'India');
Query OK, 3 rows affected (0.03 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> INSERT INTO Aircraft (model, capacity) VALUES
    -> ('Boeing 737', 180),
    -> ('Airbus A320', 150),
    -> ('Boeing 777', 300);
Query OK, 3 rows affected (0.02 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> INSERT INTO Flights (flight_number, source_airport, destination_airport,
    -> ('AI101', 1, 2, '2026-04-15 08:00:00', '2026-04-15 10:00:00', 1),
    -> ('AI202', 2, 3, '2026-04-16 09:00:00', '2026-04-16 11:30:00', 2),
    -> ('AI303', 3, 1, '2026-04-17 06:00:00', '2026-04-17 08:30:00', 3);
Query OK, 3 rows affected (0.02 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> INSERT INTO Passengers (name, email, phone) VALUES
    -> ('Rahul Sharma', 'rahul@gmail.com', '9876543210'),
    -> ('Priya Verma', 'priya@gmail.com', '9123456780'),
    -> ('Amit Singh', 'amit@gmail.com', '9988776655');
Query OK, 3 rows affected (0.02 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

6. SELECT: Retrieves data from one or more tables in a database.

```
mysql> SELECT * FROM Passengers;
+--------------+--------------+-----------------+------------+
| passenger_id | name         | email           | phone      |
+--------------+--------------+-----------------+------------+
|            1 | Rahul Sharma | rahul@gmail.com | 9876543210 |
|            2 | Priya Verma  | priya@gmail.com | 9123456780 |
|            3 | Amit Singh   | amit@gmail.com  | 9988776655 |
+--------------+--------------+-----------------+------------+
3 rows in set (0.01 sec)
```

```
mysql> SELECT name, email FROM Passengers;
+--------------+-----------------+
| name         | email           |
+--------------+-----------------+
| Rahul Sharma | rahul@gmail.com |
| Priya Verma  | priya@gmail.com |
| Amit Singh   | amit@gmail.com  |
+--------------+-----------------+
3 rows in set (0.00 sec)
```

7. UPDATE: Modifies existing records in a table based on a specified condition.

```
mysql> UPDATE Passengers
    -> SET email = 'rahul_new@gmail.com'
    -> WHERE passenger_id = 1;
Query OK, 1 row affected (0.02 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

8. DROP: Permanently deletes a database or table along with all its data and structure.

```
mysql> DROP TABLE Tickets;
Query OK, 0 rows affected (0.07 sec)
```

9. RENAME: Changes the name of an existing table in the database.

```
mysql> RENAME TABLE Passengers TO Customers;
Query OK, 0 rows affected (0.06 sec)
```

10. ALTER: Modifies the structure of an existing table, such as adding, deleting, or changing columns.

```
mysql> ALTER TABLE Customers
    -> ADD address VARCHAR(150);
Query OK, 0 rows affected (0.06 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE Customers
    -> DROP COLUMN address;
Query OK, 0 rows affected (0.05 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

11. INDEX: Improves the speed of data retrieval operations on a table by creating a fast lookup for columns.

```
mysql> CREATE INDEX idx_customer_email
    -> ON Customers (email);
Query OK, 0 rows affected (0.08 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql>
mysql> CREATE INDEX idx_customer_phone
    -> ON Customers (phone);
Query OK, 0 rows affected (0.09 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

12. SELECT with WHERE: Retrieves specific records from a table based on a given condition.

```
mysql> SELECT * FROM Customers
    -> WHERE name = 'Rahul Sharma';
+--------------+--------------+--------------------+------------+
| passenger_id | name         | email              | phone      |
+--------------+--------------+--------------------+------------+
|            1 | Rahul Sharma | rahul_new@gmail.com | 9876543210 |
+--------------+--------------+--------------------+------------+
1 row in set (0.00 sec)
```

13. DELETE: Removes specific records from a table based on a condition.

```
mysql> DELETE FROM Customers
    -> WHERE passenger_id = 1;
Query OK, 1 row affected (0.02 sec)
```

14. SELECT COUNT(*): Returns the total number of rows present in a table.

```
mysql> SELECT COUNT(*) AS total_flights
    -> FROM Flights;
+---------------+
| total_flights |
+---------------+
|             3 |
+---------------+
1 row in set (0.01 sec)
```

15. TRUNCATE: Quickly removes all records from a table while keeping its structure intact.

```
mysql> INSERT INTO SampleTable (name) VALUES
    -> ('Alpha'),
    -> ('Bravo'),
    -> ('Charlie');
Query OK, 3 rows affected (0.02 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> TRUNCATE TABLE SampleTable;
Query OK, 0 rows affected (0.09 sec)
```

Mini Project : Civic Voice Interface System

SQL Queries:

1. SHOW

```
mysql> SHOW DATABASES;
+----------------------+
| Database             |
+----------------------+
| airline              |
| civic_voice_interface |
| employee             |
| information_schema   |
| mysql                |
| performance_schema   |
| sys                  |
+----------------------+
7 rows in set (0.02 sec)
```

```
mysql> SHOW TABLES;
+--------------------------------+
| Tables_in_civic_voice_interface |
+--------------------------------+
| appuser                        |
| citizenprofile                 |
| department                     |
| governmentservice              |
| userdocuments                  |
+--------------------------------+
5 rows in set (0.01 sec)
```

2. USE

```
mysql>  USE Civic_Voice_Interface;
Database changed
```

3. CREATE

```
mysql> CREATE DATABASE Civic_Voice_Interface;
Query OK, 1 row affected (0.01 sec)
```

```
mysql>  USE Civic_Voice_Interface;
Database changed
mysql> CREATE TABLE User (
    ->     user_id INT PRIMARY KEY AUTO_INCREMENT,
    ->     full_name VARCHAR(100) NOT NULL,
    ->     email VARCHAR(100) UNIQUE NOT NULL,
    ->     phone VARCHAR(15) UNIQUE,
    ->     password_hash VARCHAR(255) NOT NULL,
    ->     role ENUM('citizen', 'admin', 'official') DEFAULT 'citizen',
    ->     created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
    -> );
Query OK, 0 rows affected (0.11 sec)

mysql> CREATE TABLE CitizenProfile (
    ->     profile_id INT PRIMARY KEY AUTO_INCREMENT,
    ->     user_id INT UNIQUE,
    ->     address TEXT,
    ->     date_of_birth DATE,
    ->     gender ENUM('male', 'female', 'other'),
    ->     aadhaar_number VARCHAR(20) UNIQUE,
    ->
    ->     FOREIGN KEY (user_id) REFERENCES User(user_id)
    -> );
Query OK, 0 rows affected (0.08 sec)
```

```
mysql> CREATE TABLE UserDocument (
    ->     document_id INT PRIMARY KEY AUTO_INCREMENT,
    ->     user_id INT,
    ->     document_type VARCHAR(50),
    ->     document_number VARCHAR(100),
    ->     file_path VARCHAR(255),
    ->     uploaded_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    ->     status ENUM('pending', 'verified', 'rejected') DEFAULT 'pending',
    ->
    ->     FOREIGN KEY (user_id) REFERENCES User(user_id)
    -> );
Query OK, 0 rows affected (0.14 sec)

mysql> CREATE TABLE Department (
    ->     department_id INT PRIMARY KEY AUTO_INCREMENT,
    ->     department_name VARCHAR(100) NOT NULL,
    ->     description TEXT,
    ->     contact_email VARCHAR(100),
    ->     contact_phone VARCHAR(15)
    -> );
Query OK, 0 rows affected (0.06 sec)

mysql> CREATE TABLE GovernmentService (
    ->     service_id INT PRIMARY KEY AUTO_INCREMENT,
    ->     department_id INT,
    ->     service_name VARCHAR(100) NOT NULL,
    ->     description TEXT,
    ->     eligibility_criteria TEXT,
    ->     processing_time VARCHAR(50),
    ->
    ->     FOREIGN KEY (department_id) REFERENCES Department(department_id)
    -> );
Query OK, 0 rows affected (0.09 sec)
```

4. DESC (DESCRIBE)

```
mysql> DESC User;
+---------------+----------------------------------+------+-----+-------------------+-------------------+
| Field         | Type                             | Null | Key | Default           | Extra             |
+---------------+----------------------------------+------+-----+-------------------+-------------------+
| user_id       | int                              | NO   | PRI | NULL              | auto_increment    |
| full_name     | varchar(100)                     | NO   |     | NULL              |                   |
| email         | varchar(100)                     | NO   | UNI | NULL              |                   |
| phone         | varchar(15)                      | YES  | UNI | NULL              |                   |
| password_hash | varchar(255)                     | NO   |     | NULL              |                   |
| role          | enum('citizen','admin','official') | YES  |     | citizen           |                   |
| created_at    | timestamp                        | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+---------------+----------------------------------+------+-----+-------------------+-------------------+
7 rows in set (0.02 sec)
```

5. INSERT

```
mysql> INSERT INTO User (user_id, full_name, email, phone, password_hash, role, created_at)
    -> VALUES
    -> (1, 'Atharva Patil', 'atharva@gmail.com', '9876543210', 'hash1', 'citizen', NOW()),
    -> (2, 'Sneha Joshi', 'sneha@gmail.com', '9123456780', 'hash2', 'citizen', NOW()),
    -> (3, 'Rajesh Kumar', 'rajesh@gov.in', '9988776655', 'hash3', 'official', NOW());
Query OK, 3 rows affected (0.03 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> INSERT INTO CitizenProfile (profile_id, user_id, address, date_of_birth, gender, aadhaar_number)
    -> VALUES
    -> (1, 1, 'Pune, Maharashtra', '2003-05-10', 'male', '123412341234'),
    -> (2, 2, 'Mumbai, Maharashtra', '2002-08-15', 'female', '567856785678'),
    -> (3, 3, 'Delhi, India', '1990-01-20', 'male', '987698769876');
Query OK, 3 rows affected (0.02 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> INSERT INTO Department (department_id, department_name, description, contact_email, contact_phone)
    -> VALUES
    -> (1, 'Water Supply', 'Manages water distribution', 'water@gov.in', '02011111111'),
    -> (2, 'Electricity', 'Handles electricity services', 'electricity@gov.in', '02022222222'),
    -> (3, 'Transport', 'Vehicle and license services', 'transport@gov.in', '02033333333');
Query OK, 3 rows affected (0.02 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

6. SELECT

```
mysql> SELECT * FROM User;
+---------+---------------+-------------------+------------+---------------+----------+---------------------+
| user_id | full_name     | email             | phone      | password_hash | role     | created_at          |
+---------+---------------+-------------------+------------+---------------+----------+---------------------+
|       1 | Atharva Patil | atharva@gmail.com | 9876543210 | hash1         | citizen  | 2026-04-14 04:56:44 |
|       2 | Sneha Joshi   | sneha@gmail.com   | 9123456780 | hash2         | citizen  | 2026-04-14 04:56:44 |
|       3 | Rajesh Kumar  | rajesh@gov.in     | 9988776655 | hash3         | official | 2026-04-14 04:56:44 |
+---------+---------------+-------------------+------------+---------------+----------+---------------------+
3 rows in set (0.00 sec)
```

```
mysql> SELECT full_name, email, role
    -> FROM User;
+---------------+-------------------+----------+
| full_name     | email             | role     |
+---------------+-------------------+----------+
| Atharva Patil | atharva@gmail.com | citizen  |
| Sneha Joshi   | sneha@gmail.com   | citizen  |
| Rajesh Kumar  | rajesh@gov.in     | official |
+---------------+-------------------+----------+
3 rows in set (0.00 sec)
```

7. UPDATE

```
mysql> UPDATE User
    -> SET phone = '9998887776'
    -> WHERE user_id = 1;
Query OK, 1 row affected (0.02 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE GovernmentService
    -> SET processing_time = '10 days'
    -> WHERE service_id = 1;
Query OK, 1 row affected (0.02 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

8. DROP

```
mysql> DROP TABLE UserDocument;
Query OK, 0 rows affected (0.06 sec)
```

9. RENAME

```
mysql> RENAME TABLE User TO AppUser;
Query OK, 0 rows affected (0.13 sec)
```

10. ALTER

```
mysql> ALTER TABLE AppUser
    -> MODIFY full_name VARCHAR(150);
Query OK, 0 rows affected (0.22 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

11. INDEX

```
mysql> CREATE INDEX idx_department
    -> ON GovernmentService (department_id);
Query OK, 0 rows affected (0.09 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

12. SELECT with WHERE

```
mysql> SELECT * FROM AppUser
    -> WHERE user_id = 1;
+---------+---------------+-------------------+------------+---------------+---------+---------------------+
| user_id | full_name     | email             | phone      | password_hash | role    | created_at          |
+---------+---------------+-------------------+------------+---------------+---------+---------------------+
|       1 | Atharva Patil | atharva@gmail.com | 9998887776 | hash1         | citizen | 2026-04-14 04:56:44 |
+---------+---------------+-------------------+------------+---------------+---------+---------------------+
1 row in set (0.00 sec)
```

13. DELETE

```
mysql> DELETE FROM GovernmentService
    -> WHERE department_id = 2;
Query OK, 1 row affected (0.01 sec)
```

14. SELECT COUNT(*)

```
mysql> SELECT COUNT(*) FROM AppUser;
+----------+
| COUNT(*) |
+----------+
|        3 |
+----------+
1 row in set (0.01 sec)
```

15. TRUNCATE

```
mysql> TRUNCATE TABLE UserDocuments;
Query OK, 0 rows affected (0.07 sec)

mysql> SELECT * FROM UserDocuments;
Empty set (0.01 sec)
```